



# Engineering Electromagnetics

## 工程电磁场

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Given the potential field  $V = 100xz/(x^2 + 4)$  V in free space: (a) Find  $\mathbf{D}$  at the surface  $z = 0$ . (b) Show that the  $z = 0$  surface is an equipotential surface. (c) Assume that the  $z = 0$  surface is a conductor and find the total charge on that portion of the conductor defined by  $0 < x < 2$ ,  $-3 < y < 0$ .

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Given the potential field

$$V = \frac{100xz}{x^2 + 4} \text{ V}$$

in free space:

a) Find  $\mathbf{D}$  at the surface  $z = 0$ : Use

$$\mathbf{E} = -\nabla V = -100z \frac{\partial}{\partial x} \left( \frac{x}{x^2 + 4} \right) \mathbf{a}_x - 0 \mathbf{a}_y - \frac{100x}{x^2 + 4} \mathbf{a}_z \text{ V/m}$$

At  $z = 0$ , we use this to find

$$\mathbf{D}(z = 0) = \epsilon_0 \mathbf{E}(z = 0) = \underline{-\frac{100\epsilon_0 x}{x^2 + 4} \mathbf{a}_z \text{ C/m}^2}$$

- b) Show that the  $z = 0$  surface is an equipotential surface: There are two reasons for this: 1)  $\mathbf{E}$  at  $z = 0$  is everywhere  $z$ -directed, and so moving a charge around on the surface involves doing no work; 2) When evaluating the given potential function at  $z = 0$ , the result is 0 for all  $x$  and  $y$ .
- c) Assume that the  $z = 0$  surface is a conductor and find the total charge on that portion of the conductor defined by  $0 < x < 2$ ,  $-3 < y < 0$ : We have

$$\rho_s = \mathbf{D} \cdot \mathbf{a}_z \Big|_{z=0} = -\frac{100\epsilon_0 x}{x^2 + 4} \text{ C/m}^2$$

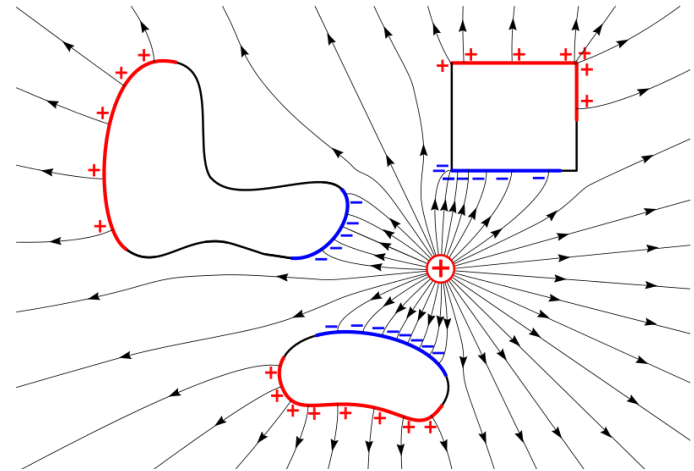
So

$$Q = \int_{-3}^0 \int_0^2 -\frac{100\epsilon_0 x}{x^2 + 4} dx dy = -(3)(100)\epsilon_0 \left( \frac{1}{2} \right) \ln(x^2 + 4) \Big|_0^2 = -150\epsilon_0 \ln 2 = \underline{-0.92 \text{ nC}}$$

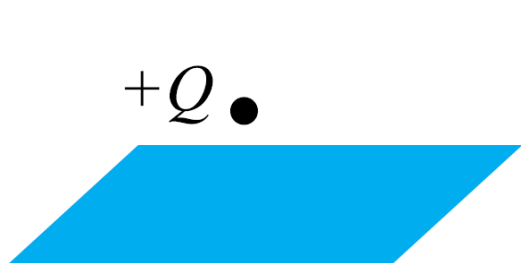
# 9: Conductors & Dielectrics

- Method of Images 镜像法

Think about the effect of metal conductors on the electric field built by the point charge in free space. Why are the streamlines distorted? How can we evaluate such kind of distortion? Can we analytically solve the distorted field?



Unfortunately, in most cases, we are not able to derive analytical solution. However, sometimes we can!



$+Q \bullet$

Conducting plane,  $V = 0$

# 9: Conductors & Dielectrics

- Method of Images 镜像法



The *Theorem of Uniqueness* 唯一性定理 states that if we are given a configuration of charges and boundary conditions, there will exist *only one potential* and electric field solution.

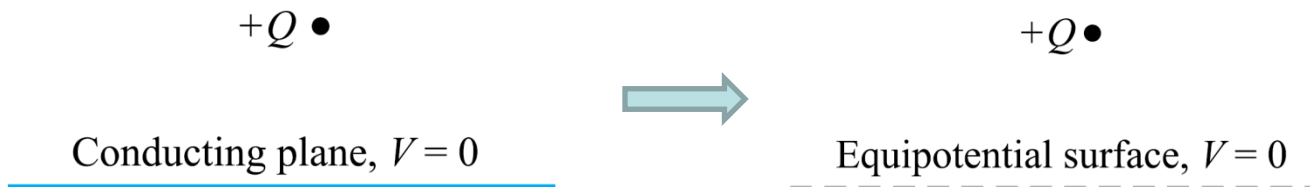


$-Q$  •

So the boundary conditions and charges are *identical in the upper half* spaces of both configurations (*not in the lower half*).

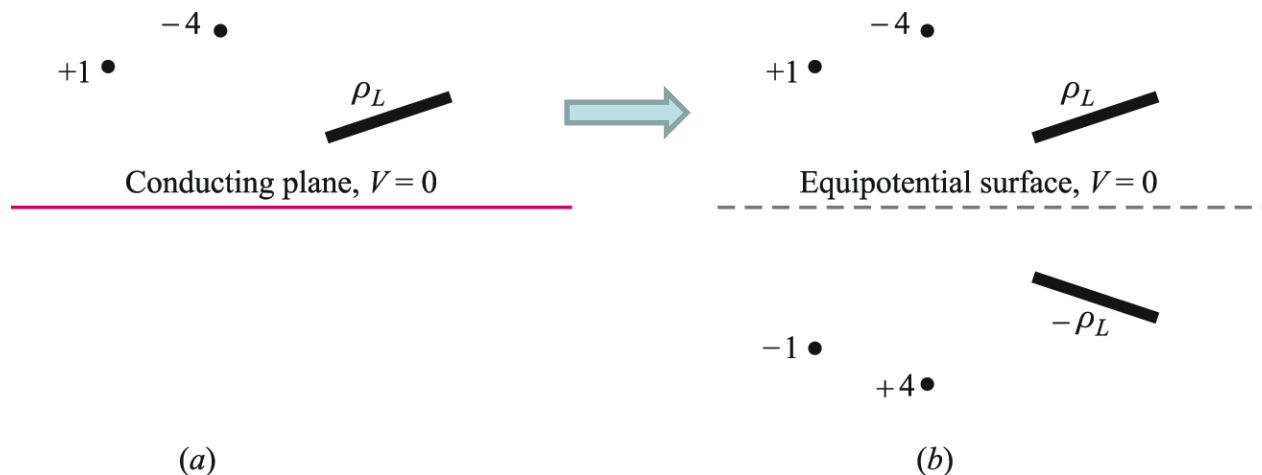
# 9: Conductors & Dielectrics

- Method of Images 镜像法



$-Q$

In effect, the positive point charge *images* across the conducting plane, allowing the conductor to be replaced by the image. The field and potential distribution in the **upper half space** is now found much more easily!

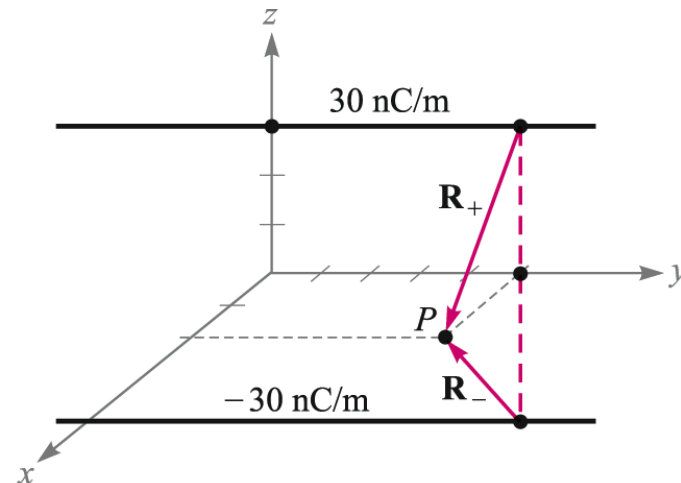
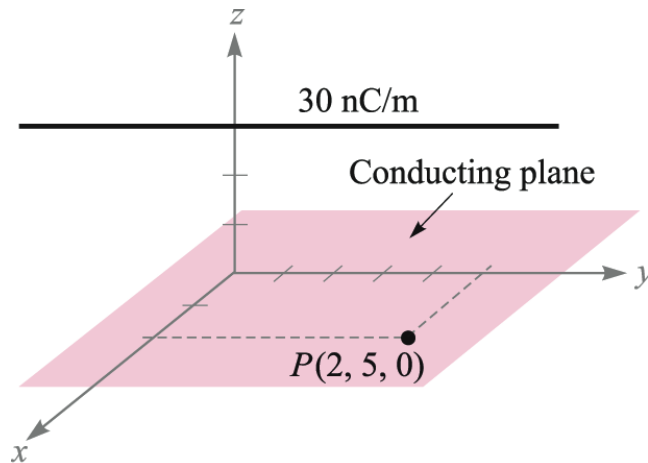


# 9: Conductors & Dielectrics

- Example

In this case, we are to find the surface charge density on the conducting plane at the point  $(2,5,0)$ . A  $30\text{-nC}$  line charge lies parallel to the  $y$  axis at  $z = 3$ .

The first step is to replace the conducting plane by a line charge of  $-30\text{ nC}$  at  $z = -3$ .



# 9: Conductors & Dielectrics

## • Example (Cont'd)

In this case, we are to find the surface charge density on the conducting plane at the point (2,5,0). A 30-nC line charge lies parallel to the y axis at  $z = 3$ .

Referring to the figure, we find:

$$\mathbf{R}_+ = 2\mathbf{a}_x - 3\mathbf{a}_z \quad \mathbf{R}_- = 2\mathbf{a}_x + 3\mathbf{a}_z$$

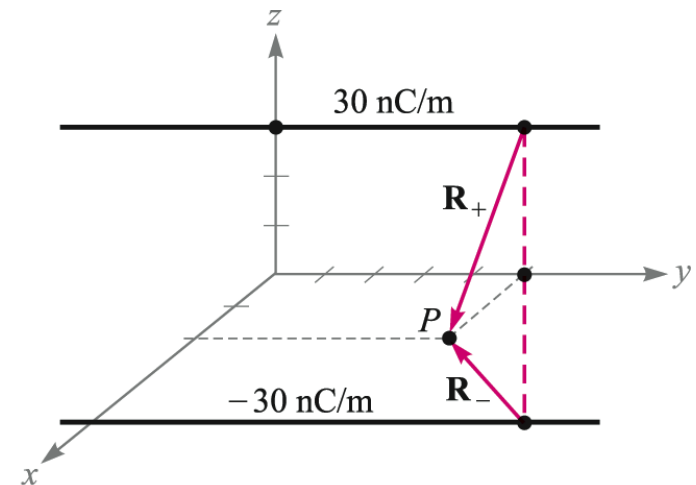
Then

$$\mathbf{E}_+ = \frac{\rho_L}{2\pi\epsilon_0 R_+} \mathbf{a}_{R_+} = \frac{30 \times 10^{-9}}{2\pi\epsilon_0 \sqrt{13}} \frac{2\mathbf{a}_x - 3\mathbf{a}_z}{\sqrt{13}}$$

$$\mathbf{E}_- = \frac{30 \times 10^{-9}}{2\pi\epsilon_0 \sqrt{13}} \frac{2\mathbf{a}_x + 3\mathbf{a}_z}{\sqrt{13}}$$

We now add the two fields to get:

$$\mathbf{E} = \frac{-180 \times 10^{-9} \mathbf{a}_z}{2\pi\epsilon_0 (13)} = -249 \mathbf{a}_z \text{ V/m}$$





# 9: Conductors & Dielectrics

- Example (Cont'd)

In this case, we are to find the surface charge density on the conducting plane at the point (2,5,0). A 30-nC line charge lies parallel to the y axis at  $z = 3$ .

We now have the electric field at point  $P$ :  $\mathbf{E} = \frac{-180 \times 10^{-9} \mathbf{a}_z}{2\pi\epsilon_0(13)} = -249 \mathbf{a}_z \text{ V/m}$

Now  $\mathbf{D} = \epsilon_0 \mathbf{E} = -2.20 \mathbf{a}_z \text{ nC/m}^2$

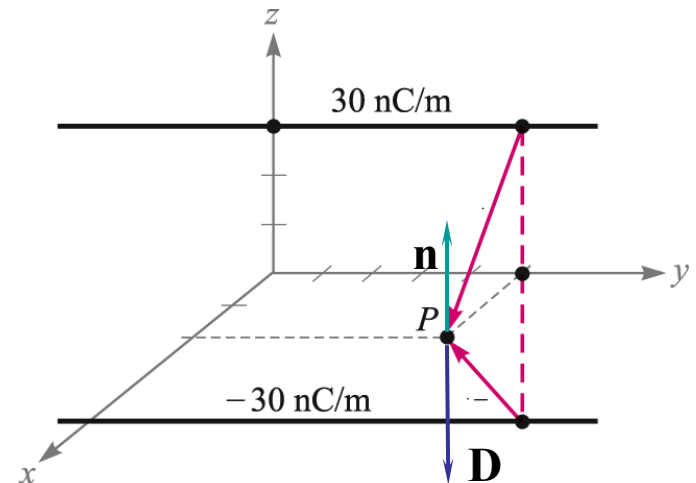
To find the charge density, use

$$\mathbf{D} \cdot \mathbf{n} \Big|_s = \rho_s$$

where  $\mathbf{n} = \mathbf{a}_z$

Therefore

$$\mathbf{D} \cdot \mathbf{n} = -2.20 \mathbf{a}_z \cdot \mathbf{a}_z = \underline{-2.20 \text{ nC/m}^2}$$





# 9: Conductors & Dielectrics

- Dielectric Materials 电介质材料

**Free space (vacuum):** There is nothing out there!

**Metal conductor:** Copper, iron, aluminum.....,

- 1) Charge can exist only on the surface;
- 2) Electric field *cannot* exist in the interior;
- 3) Conductor is an *equipotential*;
- 4) Conductor-to-Free space boundary conditions.

**Dielectric Materials (isolators):** Air, Fiber, Glass, Paper, Quartz, Porcelain 陶瓷, Polyethylene 聚乙烯

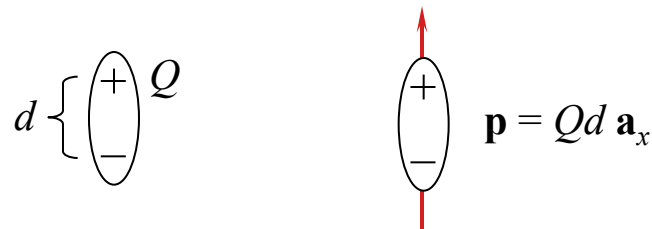
**Q:** What is the (electromagnetic) nature of dielectric materials? And Why?

# 9: Conductors & Dielectrics

- Dielectric Materials -Electric Dipole and Dipole Moment

In dielectric, charges are held in position (**bound 束缚**), and ideally there are no **free charges** that can move and form a current. Atoms and molecules may be **polar 极化** (having separated positive and negative charges) **by the application of an electric field**.

Consider such a **polarized atom or molecule**, which possesses a **dipole moment**,  $\mathbf{p}$ , defined as the charge magnitude present,  $Q$ , times the positive and negative charge separation,  $d$ . Dipole moment is a **vector** that points from the negative to the positive charge.



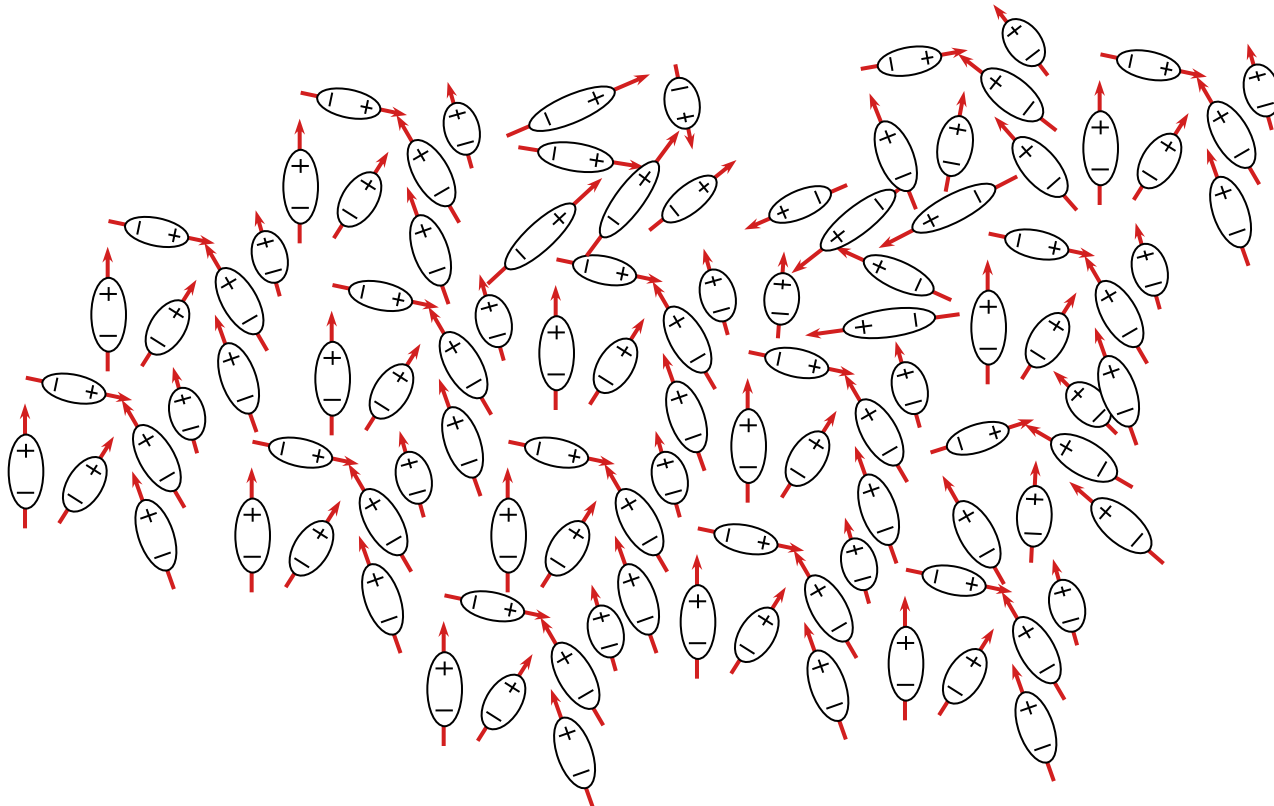
Summary-Some new terms:

Free charges, bound charges(束缚电荷), polar molecule(极性分子), non-polar molecule(无极性分子), polarize(极化).....

# 9: Conductors & Dielectrics

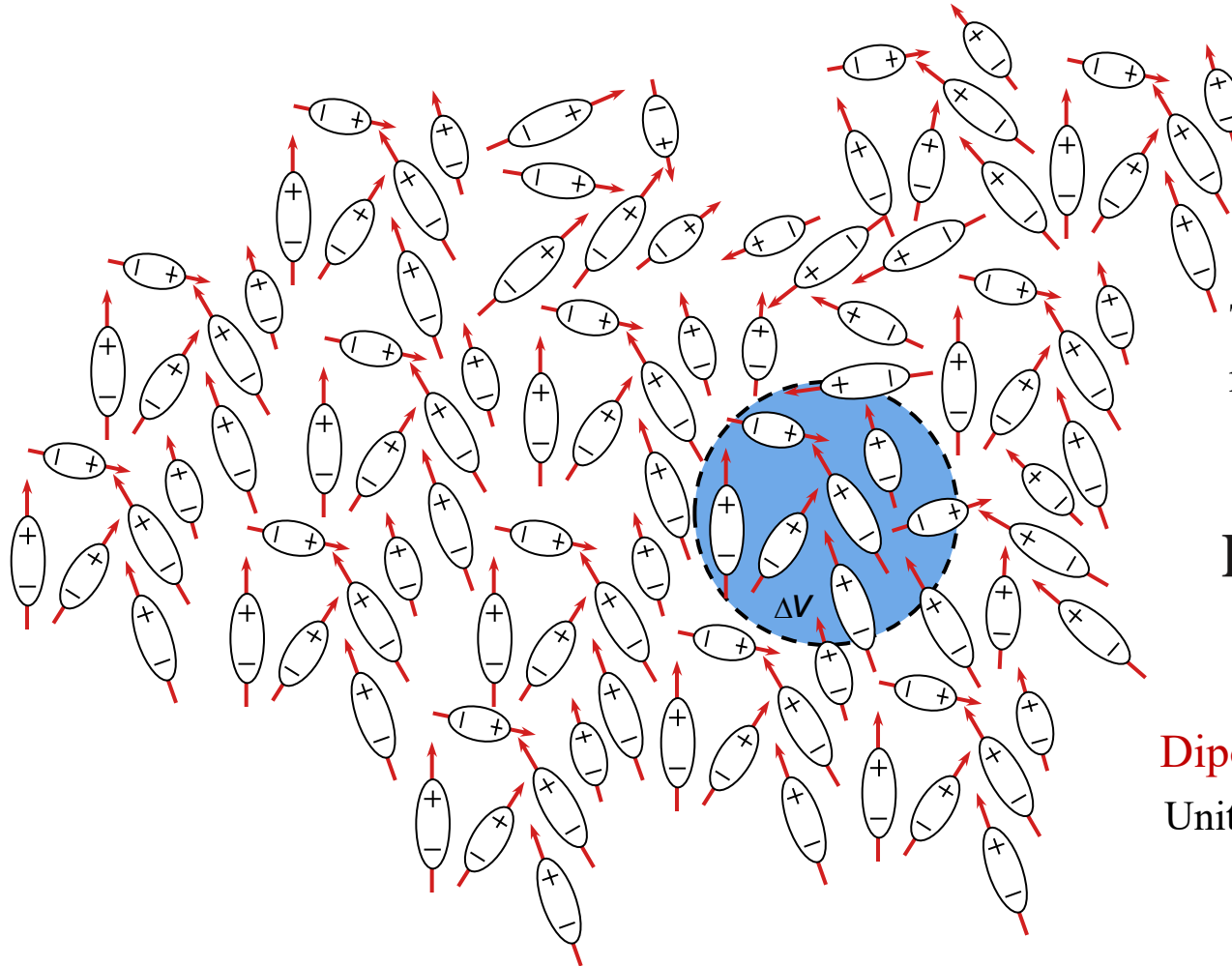
- Dielectric Materials -Model of a Dielectric

A dielectric can be modeled as an ensemble of bound charges in free space, associated with the atoms and molecules that make up the material. Some of these may have **intrinsic dipole moments**, others not. In some materials (such as liquids), dipole moments are in random directions.



# 9: Conductors & Dielectrics

- Dielectric Materials -Polarization Field



The number of dipoles is expressed as a density,  $n$  dipoles per unit volume.

The *Polarization Field* of the medium is defined as:

$$\mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{1}{\Delta v} \sum_{i=1}^{n \Delta v} \mathbf{p}_i$$

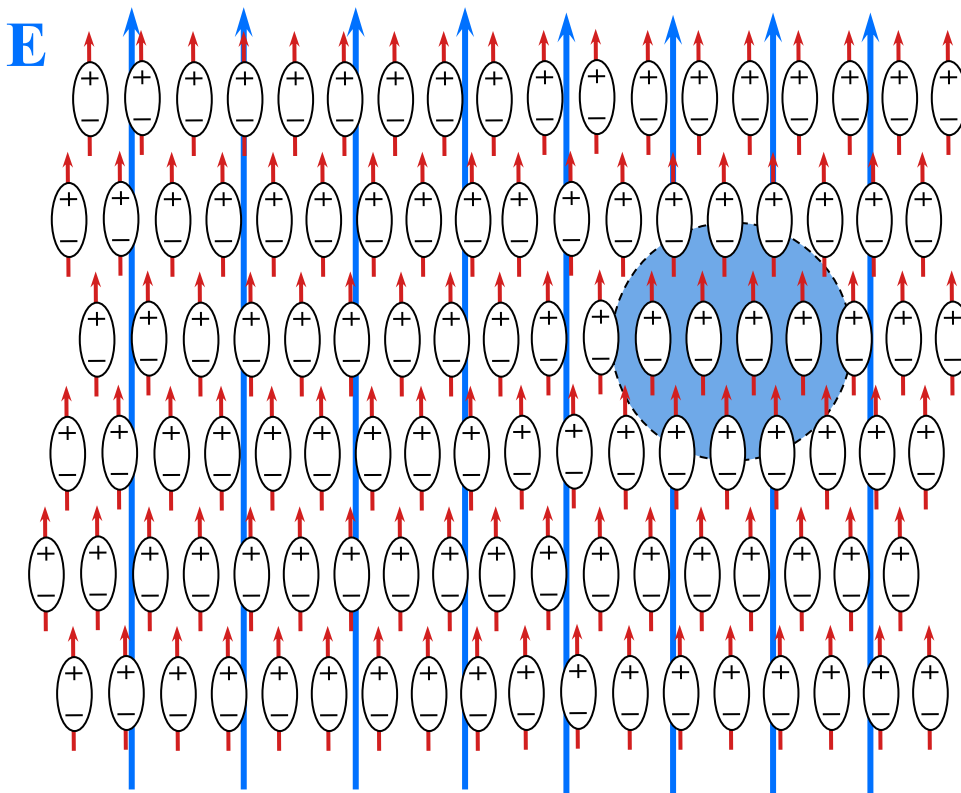
Dipole moment per unit volume

Unit: [C/m<sup>2</sup>]

# 9: Conductors & Dielectrics

- Polarization Field (with Electric Field Applied)

Introducing an electric field **may increase the charge separation in each dipole**, and **possibly re-orient dipoles** so that there is some aggregate alignment, as shown here. The effect is small, and **is greatly exaggerated here!**



The effect is to increase **P**.

$$\mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{1}{\Delta v} \sum_{i=1}^n \mathbf{p}_i$$

if all dipoles are identical

$$= n\mathbf{p}$$

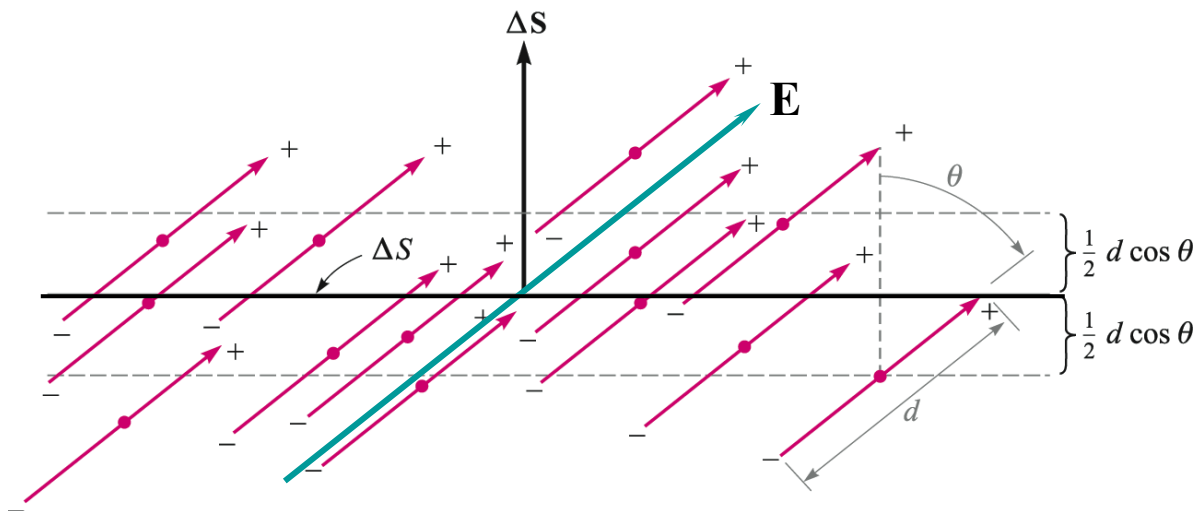
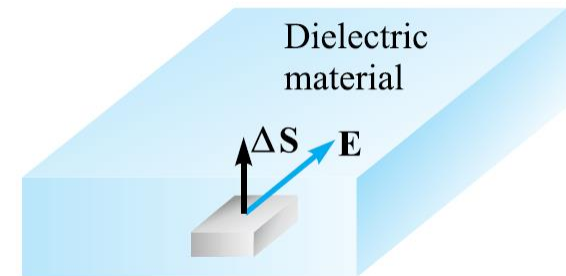
Note: There should be something to indicate the polarization degree of specific dielectric material.

# 9: Conductors & Dielectrics

- Migration of Bound Charge

Consider an electric field applied at an angle  $\theta$  to a surface normal as shown. The resulting separation of bound charges (or re-orientation) **leads to positive bound charge crossing upward through surface of area  $\Delta S$ , while negative bound charge crosses downward through the surface.**

Dipole centers (red dots) that lie within the range  $(1/2) d \cos \theta$  above or below the surface will transfer charge across the surface.

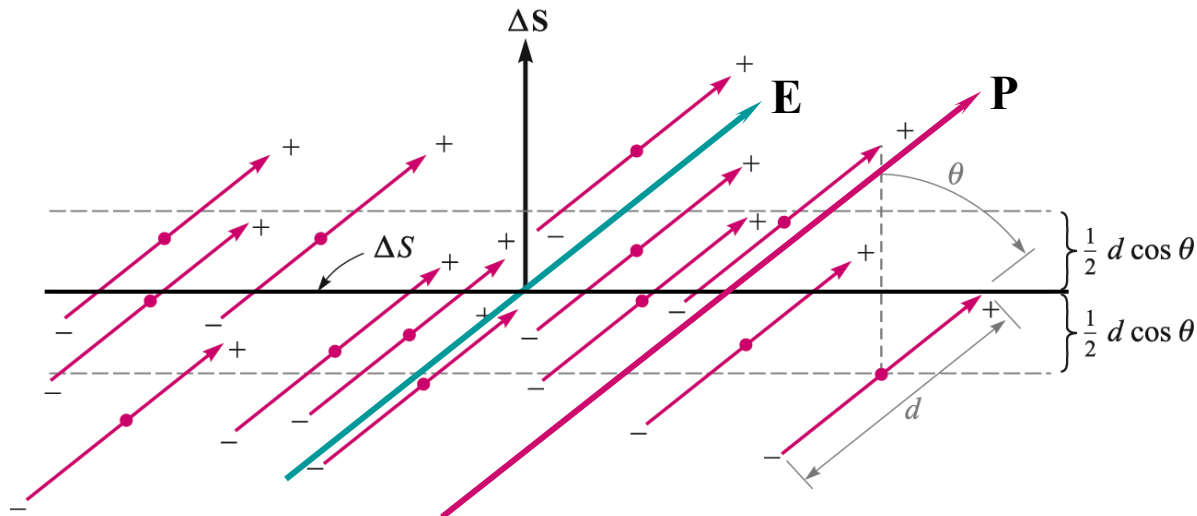


# 9: Conductors & Dielectrics

- Migration of Bound Charge

The total bound charge that crosses the surface is given by:

$$\Delta Q_b = n Q \underbrace{d \cos \theta \Delta S}_{\text{volume}} = n Q \mathbf{d} \cdot \Delta \mathbf{S} = \underline{\mathbf{P} \cdot \Delta \mathbf{S}}$$





# 9: Conductors & Dielectrics

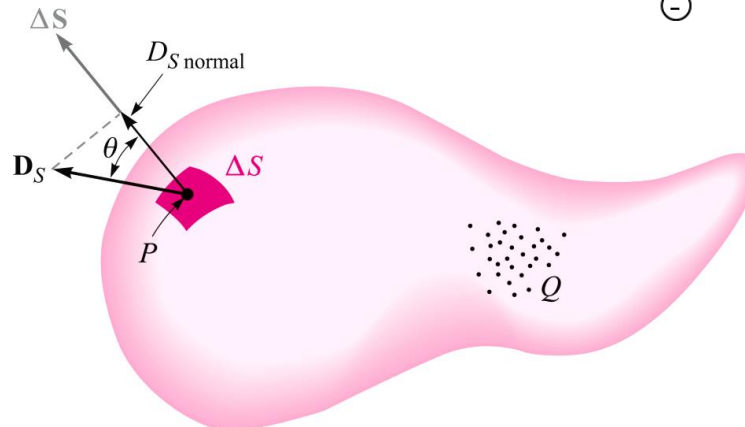
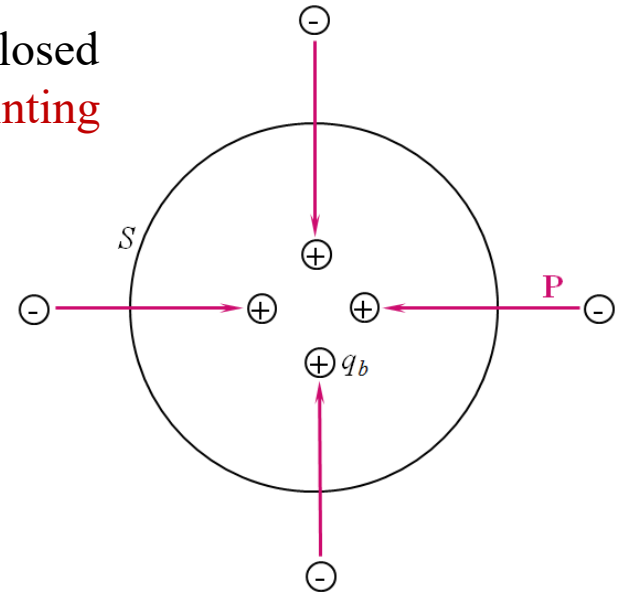
- Polarization Flux Through a Closed Surface

The accumulation of positive bound charge within a closed surface means that the polarization vector must be **pointing inward**. Therefore:

$$Q_b = - \oint_S \mathbf{P} \cdot d\mathbf{S}$$

Gauss's Law in free space:

$$Q = \oint_S \mathbf{D}_S \cdot d\mathbf{S}$$



# 9: Conductors & Dielectrics

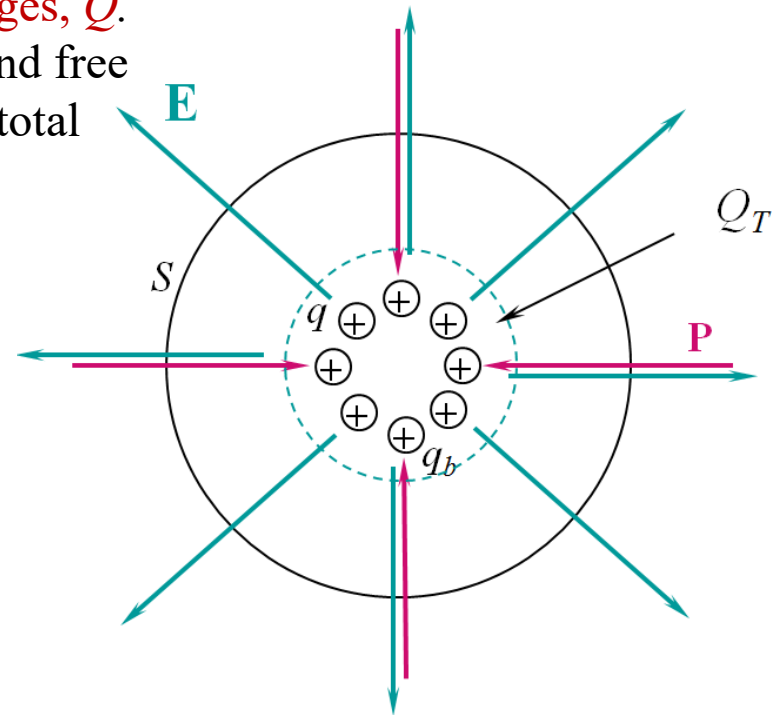
- Polarization Flux Through a Closed Surface

Now consider the charge within the closed surface consisting of **bound charges,  $Q_b$** , and **free charges,  $Q$** . The total charge will be the sum of all bound and free charges. We write Gauss' Law in terms of the total charge,  $Q_T$  as:

$$Q_T = \oint_S \epsilon_0 \mathbf{E} \cdot d\mathbf{S}$$

where

$$Q_T = \underset{\substack{\uparrow \\ \text{bound charge}}}{Q_b} + \underset{\substack{\downarrow \\ \text{free charge}}}{Q}$$



# 9: Conductors & Dielectrics

- Polarization Flux Through a Closed Surface

We now have:

$$Q_b = - \oint_S \mathbf{P} \cdot d\mathbf{S} \quad \text{and} \quad Q_T = \oint_S \epsilon_0 \mathbf{E} \cdot d\mathbf{S}$$

where  $Q_T = Q_b + Q$

combining these, we write:

$$Q = Q_T - Q_b = \oint_S (\epsilon_0 \mathbf{E} + \mathbf{P}) \cdot d\mathbf{S}$$

we thus identify:

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

which we use in the familiar form of Gauss' Law:

$$Q = \oint_S \mathbf{D} \cdot d\mathbf{S}$$

# 9: Conductors & Dielectrics

- Charge Densities

Taking the previous results and using **the divergence theorem**, we find the point form expressions:

$$\text{Bound Charge: } Q_b = \int_v \rho_b dv = - \oint_S \mathbf{P} \cdot d\mathbf{S} \longrightarrow \boxed{\nabla \cdot \mathbf{P} = -\rho_b}$$

$$\text{Total Charge: } Q_T = \int_v \rho_T dv = \oint_S \epsilon_0 \mathbf{E} \cdot d\mathbf{S} \longrightarrow \boxed{\nabla \cdot \epsilon_0 \mathbf{E} = \rho_T}$$

$$\text{Free Charge: } Q = \int_v \rho_v dv = \oint_S \mathbf{D} \cdot d\mathbf{S} \longrightarrow \boxed{\nabla \cdot \mathbf{D} = \rho_v}$$



# 9: Conductors & Dielectrics

- Electric Susceptibility 电极化率 and the relative permittivity 相对介电常数

A stronger electric field results in a larger polarization in the medium. In a *linear medium*, the relation between  $\mathbf{P}$  and  $\mathbf{E}$  is *linear*, and is given by:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

where  $\chi_e$  is the electric *susceptibility* (电极化率) of the medium.

We may now write:  $\mathbf{D} = \epsilon_0 \mathbf{E} + \chi_e \epsilon_0 \mathbf{E} = (\chi_e + 1) \epsilon_0 \mathbf{E}$

where the relative permittivity (相对介电常数) is defined as:

$$\epsilon_r = \chi_e + 1$$

Leading to the overall permittivity of the medium:

$$\mathbf{D} = \epsilon \mathbf{E} \quad \text{where} \quad \epsilon = \epsilon_r \epsilon_0$$

# 9: Conductors & Dielectrics

- Isotropic各向同性 vs. Anisotropic各向异性 Media

In an *isotropic* medium, the dielectric constant is invariant with *direction* of the applied electric field.

This is not the case in an *anisotropic medium* (usually a crystal) in which the dielectric constant will *vary as the electric field is rotated in certain directions*. In this case, the electric flux density vector components must be evaluated separately through the *dielectric tensor*. The relation can be expressed in the form:

$$D_x = \epsilon_{xx} E_x + \epsilon_{xy} E_y + \epsilon_{xz} E_z$$

$$D_y = \epsilon_{yx} E_x + \epsilon_{yy} E_y + \epsilon_{yz} E_z$$

$$D_z = \epsilon_{zx} E_x + \epsilon_{zy} E_y + \epsilon_{zz} E_z$$

Note: 1) **D** and **E** (and **P**) are no longer parallel;  
2) There may be hysteresis 滞后 between **D** and **E**.



# 9: Conductors & Dielectrics

- Summary

In summary, then, we now have a relationship between  $\mathbf{D}$  and  $\mathbf{E}$  that depends on the dielectric material present,

$$\mathbf{D} = \epsilon \mathbf{E}$$

where

$$\epsilon = \epsilon_0 \epsilon_r$$

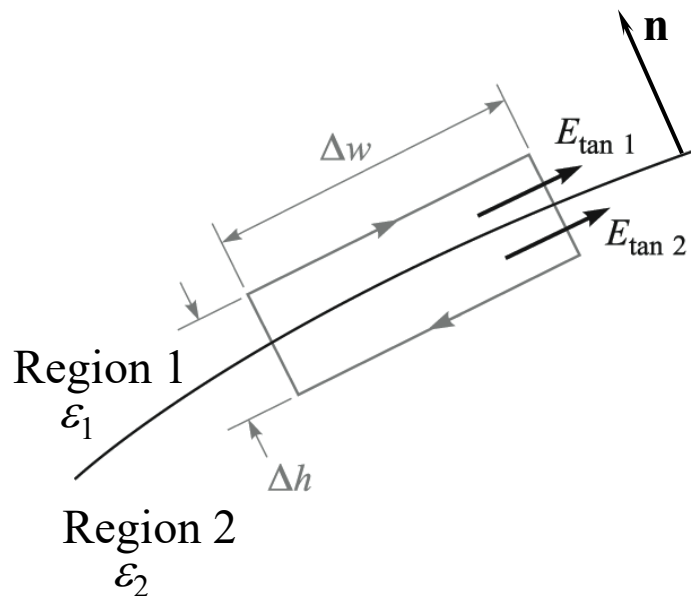
This electric flux density is still related to the free charge by either the point or integral form of Gauss's law:

$$\nabla \cdot \mathbf{D} = \rho_v$$

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q$$

# 9: Conductors & Dielectrics

- Boundary Condition for Tangential Electric Field



We use the fact that  $\mathbf{E}$  is conservative:

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

Therefore:

$$E_{\tan 1} \Delta w - E_{\tan 2} \Delta w = 0$$

Leading to:  $E_{\tan 1} = E_{\tan 2}$

More formally:

$$(\mathbf{E}_1 - \mathbf{E}_2) \times \mathbf{n} = 0$$

$$\frac{D_{\tan 1}}{\epsilon_1} = E_{\tan 1} = E_{\tan 2} = \frac{D_{\tan 2}}{\epsilon_2}$$

$$\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{\epsilon_1}{\epsilon_2}$$

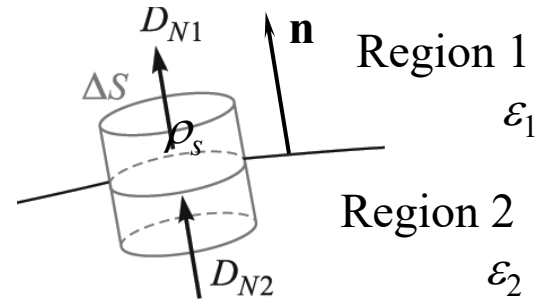


# 9: Conductors & Dielectrics

- Boundary Condition for Normal Electric Flux Density

We apply Gauss' Law to the cylindrical volume shown here, in which cylinder height is allowed to approach zero, and there is charge density  $\rho_s$  on the surface:

$$Q = \oint_S \mathbf{D} \cdot d\mathbf{S}$$



The electric flux enters and exits only through the bottom and top surfaces, respectively.

$$D_{N1} \Delta S - D_{N2} \Delta S = \Delta Q = \rho_s \Delta S$$

From which:

$$D_{N1} - D_{N2} = \rho_s$$

More formally:

$$(\mathbf{D}_1 - \mathbf{D}_2) \cdot \mathbf{n} = \rho_s$$

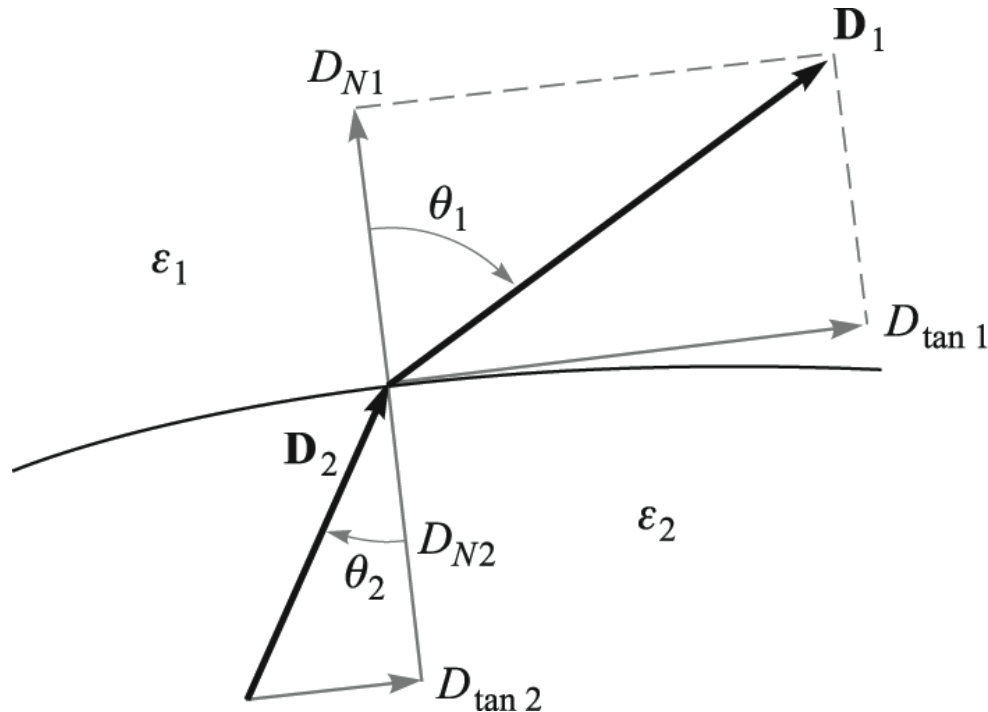
and if the charge density is zero:

$$D_{N1} = D_{N2}$$

# 9: Conductors & Dielectrics

- Example

We wish to find the relation between the angles  $\theta_1$  and  $\theta_2$ , **assuming no charge density on the surface.**



# 9: Conductors & Dielectrics

- Example

We wish to find the relation between the angles  $\theta_1$  and  $\theta_2$ , **assuming no charge density on the surface**.

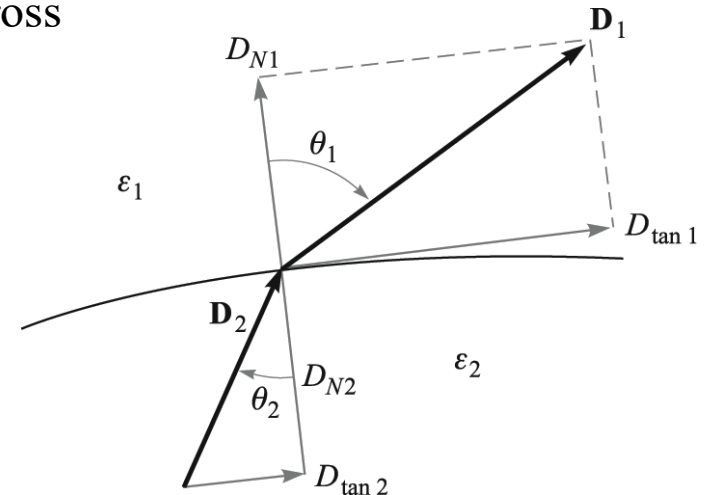
The normal components of  $\mathbf{D}$  will be continuous across the boundary, so that:

$$\underline{D_{N1} = D_1 \cos \theta_1 = D_2 \cos \theta_2 = D_{N2}}$$

Then, with tangential  $\mathbf{E}$  continuous across the boundary, it follows that:

$$\underline{\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{D_1 \sin \theta_1}{D_2 \sin \theta_2} = \frac{\epsilon_1}{\epsilon_2}}$$

$$\text{or... } \underline{\epsilon_2 D_1 \sin \theta_1 = \epsilon_1 D_2 \sin \theta_2}$$



Now, taking the ratio of the two underlined equations, we finally obtain:

$$\boxed{\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}}$$

# HW-5



2. Given the current density  $\mathbf{J} = -10^4 [\sin(2x)e^{-2y}\mathbf{a}_x + \cos(2x)e^{-2y}\mathbf{a}_y]$  kA/m<sup>2</sup>: a)

Find the total current crossing the plane  $y = 1$  in the  $a_y$  direction in the region  $0 < x < 1$ ,  $0 < z < 2$ ; b) Find the total current leaving the region  $0 < x < 1$ ,  $0 < y < 1$ ,  $2 < z < 3$  by integrating  $\mathbf{J} \cdot d\mathbf{S}$  over the surface of the cube; c) Repeat part b, but use the divergence theorem.

4. Two perfectly-conducting cylindrical surfaces of length  $l$  are located at  $\rho = 3$  cm and  $\rho = 5$  cm. The total current passing radially outward through the medium between the cylinders is 3 A dc (direct current, 直流电流). Find the voltage and resistance between the cylinders, and  $\mathbf{E}$  in the region between the cylinders, if a conducting material having  $\sigma = 0.05$  S/m is present for  $3 < \rho < 5$  cm.
5. Two point charges of  $-100\pi$   $\mu\text{C}$  are located at  $(2, -1, 0)$  and  $(2, 1, 0)$ . The surface  $x=0$  is a conducting plane. a) Determine the surface charge density at the origin; b) Determine  $\rho_s$  at  $P(0, h, 0)$ .
6. Atomic hydrogen contains  $5.5 \times 10^{25}$  atoms/m<sup>3</sup> at a certain temperature and pressure. When an electric field of 4 kV/m is applied, each dipole formed by the electron and positive nucleus has an effective length of  $7.1 \times 10^{-19}$  m. a) Find  $\mathbf{P}$  (Dipole moment per unit volume); b) Find the relative permittivity  $\epsilon_r$ . 提示: 氢原子含有一个质子和一个电子。质子、电子的电荷量在第二次课件中可以找到。